PANOS CONSTANTINIDES

Electrical and Computer Engineering Student

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Homepage   LinkedIn

EXPERIENCE

Research Assistant - Quantum Applications

TU Crete

September 2025 - present

Working on **simulation of nonlinear dynamics and error mitigation** to simulate fluid mechanics in a scalable manner, as part of Prof. Dimitris Angelakis group. Contributing to the development of work packages in Qiskit and benchmarking.

Local Quantum Tech Student Group

IEEE TUC SB

Initiated TUC-QT, the local student group for quantum technologies under the IEEE TUC student branch. The goal of the group is to connect students across the country that find interest interested in the field, by organizing study groups and workshops.

CONFERENCES & EVENTS

- Quantum Flagship - QCFD 4th Coordination Meeting, 3/2026

Poster: Error Mitigation in Deep Parameterized Quantum Circuits

- Quantum Algorithms and Optimization in Athens, 12/2025
- IQOQI Summer School, Vienna & Innsbruck, Austria 9/2024
- Qiskit Global Summer School, 7/2025 - *excellence*
- MIT iQuHack, 1/2025
- Pennylane QHack, 1/2024
- Qiskit Global Summer School, 7/2023

QUALIFICATIONS

- Hands-on quantum error correction **(92/100)** - *Google Quantum AI*
- Variational Algorithm Design **(pass)** - *IBM*
- Introduction to Quantum Mechanics **(pass)** - *Mathesis*

TOOLS

Python	●●●●●
Git	●●●●○
Julia	●●○○○
C/C++	●●●○○
MPI+OpenMP	●●●○○
Linux	●●●●○
LaTeX	●●●○○

LANGUAGES

English	C2 Proficiency
Greek	Native speaker

EDUCATION

BSc (honors) in Electrical and Computer Engineering

TU Crete - top 4% of class

2022 present   Chania, Greece

Selected coursework
(most with high distinction)

- Quantum Technologies
- Quantum Information and Estimation Theory
- Introduction to Quantum Computation
- Statistical Modeling and Pattern Recognition
- Parallel Scientific Computing

Selected (open-source) projects

- VQE Applied on Quantum Magnetism (GitHub)
- Quantum Information Processing with Trapped Ions (GitHub)
- Quantum Information Aspects of Modified Jaynes-Cummings Models (GitHub)
- Machine LearningGuided ADAPT-VQE: Towards Problem-Agnostic Variational Quantum Simulations.

RESEARCH INTERESTS

Quantum Applications : Optimizing existing algorithms to reduce hardware requirements. Studying the intersection of Hadamard-test schemes, mid-circuit measurements with feed-forward and tensor network-inspired algorithms, while providing resource estimates for compilation on specific architectures.

Digital Quantum simulations: I am interested in digital Hamiltonian simulation and ground state estimation. Specifically algorithmic design with block encoding and linear combination of unitaries. Quantum signal processing as a unifying framework of algorithmic primitives.

Quantum Error Mitigation: Exploiting symmetries and model constraints in order to tackle errors. Improve existing methods in order to reduce bias. Conducting scalability comparisons and providing resource estimates for target applications.